Status	Finished
Started	Tuesday, 20 October 2026, 3:39 PM
Completed	Tuesday, 20 October 2026, 3:42 PM
Duration	3 mins 18 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Complete

Mark 1.00 out of 1.00

Which built-in Python function is used to sort data?

- a. arrange()
- b. sorted()
- c. order()
- d. sort()

The correct answer is: sorted()

Question **2**

Complete

Mark 1.00 out of 1.00

What is a significant characteristic of Bubble Sort?

- a. It recursively sorts subproblems
- b. It uses the heap data structure
- c. It divides the list into sublists
- d. It bubbles up the largest element in each pass

The correct answer is: It bubbles up the largest element in each pass

Question **3**

Complete

Mark 1.00 out of 1.00

_____is putting an element in the appropriate place in a sorted list yields a larger sorted order list.

- a. Insertion
- b. Selection
- c. Distribution
- d. Extraction

The correct answer is:
Insertion

Question **4**

Complete

Mark 1.00 out of 1.00

Algorithm design technique used in merge sort algorithm is

- a. Greedy method
- b. Backtracking
- c. Divide and conquer
- d. Dynamic programming

The correct answer is:
Divide and conquer

Question **5**

Complete

Mark 1.00 out of 1.00

Why is it advantageous to sort data before performing duplicate analysis?

- a. It has no effect on the analysis process
- b. It complicates the analysis process
- c. It allows for quicker identification of duplicates
- d. It makes the analysis slower

The correct answer is: It allows for quicker identification of duplicates

Question **6**

Complete

Mark 1.00 out of 1.00

Which of the following is a key reason for the importance of sorting algorithms?

- a. Sorting makes it harder to search for items
- b. Sorting is rarely used in programming
- c. Sorting decreases the efficiency of selection operations
- d. Sorting helps in finding duplicates quickly

The correct answer is: Sorting helps in finding duplicates quickly

Question **7**

Complete

Mark 1.00 out of 1.00

What type of problems can sorting help solve efficiently?

- a. Searching
- b. All of the above
- c. Duplicates
- d. Selection

The correct answer is: All of the above

Question **8**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm involves comparing elements and swapping adjacent items that are out of order?

a. Linear Search
b. Merge Sort
c. Bubble Sort
d. Binary Search

The correct answer is: Bubble Sort

Question **9**

Complete

Mark 1.00 out of 1.00

Which of the following best describes the process of Merge Sort?

a. It repeatedly finds the minimum element and moves it to the sorted part of the list
b. It compares adjacent elements and swaps them if necessary
c. It builds a sorted array one element at a time
d. It divides the list into two halves, sorts each half, and then merges them

The correct answer is: It divides the list into two halves, sorts each half, and then merges them

Question **10**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm is known for its simplicity and sequential comparison of elements?

a. Merge Sort
b. Quick Sort
c. Bubble Sort
d. Heap Sort

The correct answer is: Bubble Sort

Question **11**

Complete

Mark 1.00 out of 1.00

What is one advantage of sorting a list before performing a search operation?

a. It increases the number of comparisons needed
b. It allows for faster searching
c. It has no effect on the search operation
d. It makes the search operation slower

The correct answer is: It allows for faster searching

Question **12**

Complete

Mark 1.00 out of 1.00

Which Python function would you use to sort a list in-place?

- a. arrange()
- b. sorted()
- c. sort()
- d. order()

The correct answer is: sort()

Question **13**

Complete

Mark 1.00 out of 1.00

Which of the following is not an in-place sorting algorithm?

- a. Heap sort
- b. Quick sort
- c. Selection sort
- d. Merge sort

The correct answer is:
Merge sort

Question **14**

Complete

Mark 1.00 out of 1.00

How does Merge Sort achieve its efficiency?

- a. By using the bubble-up method
- b. By breaking the input into smaller parts and merging them
- c. By sorting data in a single pass
- d. By comparing elements sequentially

The correct answer is: By breaking the input into smaller parts and merging them

Question **15**

Complete

Mark 1.00 out of 1.00

What is one of the key advantages of using the built-in sorted() function in Python?

- a. It sorts data out of the box efficiently
- b. It is less efficient than custom sorting algorithms
- c. It only works with integer arrays
- d. It requires external libraries

The correct answer is: It sorts data out of the box efficiently